FastSpeech 2 Text-to-Speech: English (Svarah) and Telugu (IndicTTS)

Technical Report. Manya Singh. 07-11-2026. Repository: https://github.com/manyasiingh/FastSpeech_Implementation

1. Objective and summary

Two separate FastSpeech 2 models were trained from the public PyTorch implementation (ming024/FastSpeech2): English on Svarah and Telugu on IndicTTS Telugu, each on about 6 h of raw audio (about 5 h train, 1 h validation). Both ran for 30,000 steps; all results use the final step-30,000 checkpoint, not the minimum-validation checkpoint. Listening to saved checkpoints showed that English sounds best near step 10,000 and Telugu improves with training (Section 5).

	English	Telugu
Speakers / utterances	114 speaker proxies; 3,301 train, 664 val	1 female speaker; 1,547 train, 311 val
Final val total (step 30k)	3.3666 (mel 0.969, pitch 0.532, energy 0.658, dur 0.239)	4.8623 (mel 1.085, pitch 1.566, energy 0.952, dur 0.174)
Minimum val total	3.1089 at step 3,500	4.2522 at step 1,000
Final train total	1.0772	0.6965
Training time (Kaggle T4)	3 h 55 min (2.13 steps/s)	6 h 06 min (1.37 steps/s)

2. Implementation approach

The reference model was reused without architectural change, following Ren et al. (ICLR 2021):

- **Encoder/decoder:** feed-forward Transformer; 4 encoder and 6 decoder layers, hidden size 256, 2 heads, conv filter 1024 (kernels 9 and 1).
- **Variance adaptor:** duration (log domain), pitch and energy predictors (1D conv + layer norm + dropout). Pitch and energy are phoneme-level, quantised to 256 bins and embedded; a length regulator expands to mel-frame rate.
- **Loss and training:** mel and post-net L1 plus duration, pitch, energy MSE; warm-up and step-annealed learning rate. **Vocoder:** pretrained universal HiFi-GAN for both languages, not fine-tuned.
- Parameters: English 35,188,545 (multi-speaker); Telugu 35,527,233 (single-speaker).

Modifications:

Change	Language	Reason
np.int/np.float replaced; torch.load weights_only=False; keyword-argument librosa calls; energy targets cast to float32	both	numpy 2.x, PyTorch 2.6+, new librosa, dtype error
Silence handling in preprocessor.py (English: empty intervals added to silence list; Telugu: empty interval renamed sp)	both	MFA 3 writes silences as empty text
One config.yaml per language; new prepare_data.py, make_config.py, plot_losses.py; inference.py replaces synthesize.py	both	assignment structure, subsampling, plots
Speaker proxy from Svarah demographic columns; LibriTTS multi-speaker base config; LibriSpeech lexicon + g2p_en	English	no speaker id in Svarah
Telugu phone set (text/telugu_phones.json) added to symbols.py; LJSpeech base config; max_seq_len 1000 to 1800	Telugu	Telugu units; clips up to 20 s
AI4Bharat IndicMFA Telugu aligner; generated word-level dictionary; longest-match grapheme rule reused at inference	Telugu	no Telugu word dictionary

In short: the architecture is untouched. All changes are compatibility fixes, per-language configs and frontends. The Telugu-specific work is the aligner, generated dictionary and phone set, because no Telugu lexicon was available.

3. Data, preprocessing and training configuration

	English (Svarah)	Telugu (IndicTTS)
Source	ai4bharat/Svarah, 6,656 utterances; one split, no speaker id	SPRINGLab/IndicTTS_Telugu, 8,576 utterances; female speaker only (4,043 after text filtering), chosen for stability on about 6 h

	English (Svarah)	Telugu (IndicTTS)
Speaker	114 proxies from gender, age, language, state/district, qualification, job; proxies with fewer than 5 utterances removed. Approximates identity	Single speaker
Filtering	1.5 to 11 s; ASCII letters and apostrophes only; transcripts with digits dropped	1.5 to 20 s; Telugu Unicode block (U+0C00 to U+0C7F) and spaces only; any Latin letters or digits dropped
Subsampling	Shuffle (seed 42), accumulate up to 6.0 h; filters left 3,984 utterances, 5.65 h, all used. 22,050 Hz mono	Shuffle (seed 42) to 6.00 h: 1,864 utterances. The 11 s cap gave only 1.88 h, so the limit was raised to 20 s
Split	664 val, rest train, by utterance (speakers shared). Overlap check: 0.	311 val, rest train. Overlap check: 0.
Alignment	MFA 3.4.3, english_us_arpa; 3,978 of 3,984 aligned (6 dropped); no missing phones	IndicMFA Telugu (MFA 3.4.3, 362 units) plus generated dictionary for 11,619 missing word types; 1,859 of 1,864 aligned.
Features	22.05 kHz, FFT/window 1024, hop 256, 80 mel bins; PyWorld pitch and energy at phoneme level, normalised with training statistics	same
Inference frontend	Lexicon + g2p_en; hand-normalised numerals in E4	Dictionary + longest-match rule; hand-written number words; test text spacing normalised (PDF copy had spurious spaces), no characters changed
Training	Batch 16; 2,000 warm-up steps; LR x 0.3 at 20k and 26k; 30,000 steps; validate every 500, checkpoint every 2,500; max_seq_len 1000	same, max_seq_len 1800

In short: both models use about 6 h of seeded, utterance-level subsampled raw data. English is multi-speaker, accented and noisy; Telugu is one clean speaker, so the setups differ in more than language. 4.35 h for English and 3.82 h for Telugu after silence trimming and dropped utterances were used.

4. Training and validation loss analysis

English. Training total fell from 9.05 (step 100) to 1.36 (10k), 1.28 (20k) and 1.08 (30k). Pitch, energy and duration curves are noisier than mel, with one isolated pitch spike near step 3,300 that recovered; there was no divergence. Validation total bottomed at 3.1089 at step 3,500 and rose slowly to 3.3420 (20k) and 3.3666 (30k), 8.3% above the minimum (mel +5.9%, pitch +9.5%, energy +13.9%, duration +15.9%). The widening train/validation gap is consistent with overfitting after roughly 4,000 to 5,000 steps. Probable contributors are small data, a 35M-parameter model, accent and speaker variability, and no early stopping.

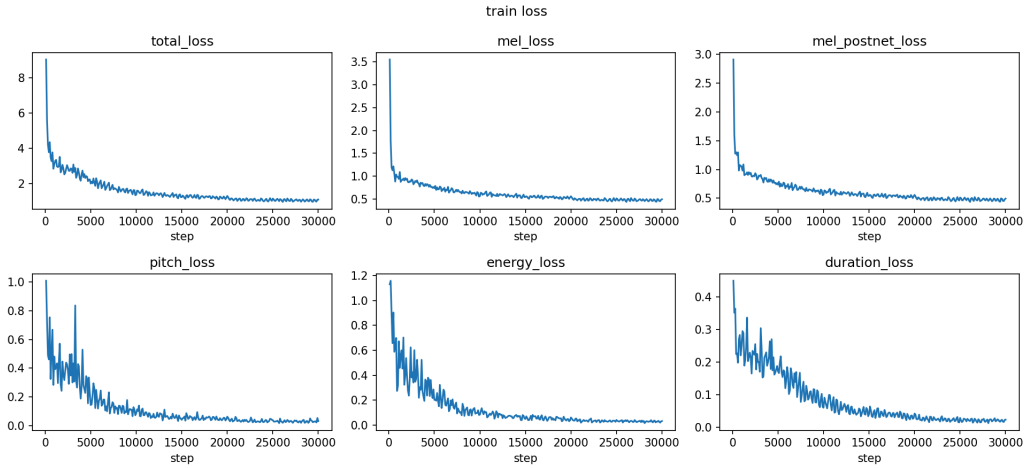


Fig.1. Training loss for English

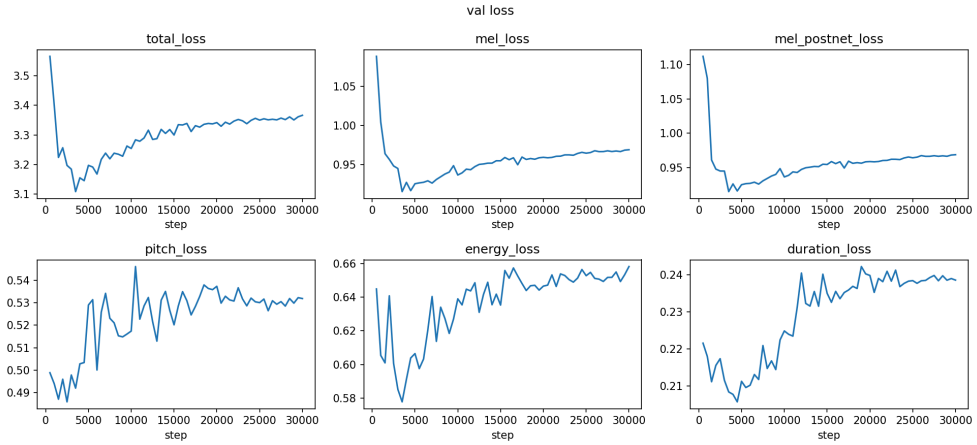


Fig.2. Validation Loss for English

Telugu. Training total fell from 11.06 (step 100) to 2.77 (1k), 1.21 (5k), 0.91 (10k), 0.78 (20k) and 0.70 (30k). Validation total was lowest at 4.2522 at step 1,000, rose to 4.84 by step 18,000 and then plateaued between 4.85 and 4.87 (4.8545 to 4.8676 over the last 2,000 steps). Pitch is the largest validation component, followed by mel and energy. This is a large generalisation gap with a late plateau, consistent with overfitting but not a late divergence. Despite the rising validation loss, the audio did not degrade: later checkpoints sounded better from step 5,000 onward (Section 5), so validation loss was a poor guide to Telugu audio quality.

English and Telugu losses are not directly comparable (different data, speaker conditions and phone inventories). Telugu is higher on mel, pitch and energy and lower on duration.

In short: both models fit the training set much better than the validation set, and validation was best very early (steps 3,500 and 1,000). Audio quality follows a different trend: English is best near step 10,000 and degrades later, while Telugu keeps improving to the reported step-30,000 checkpoint.

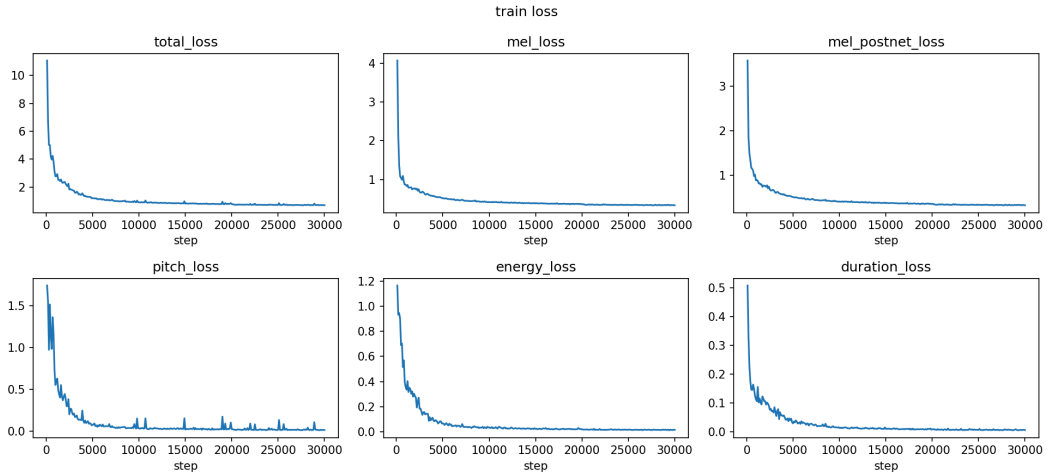


Fig.3. Training loss for Telugu

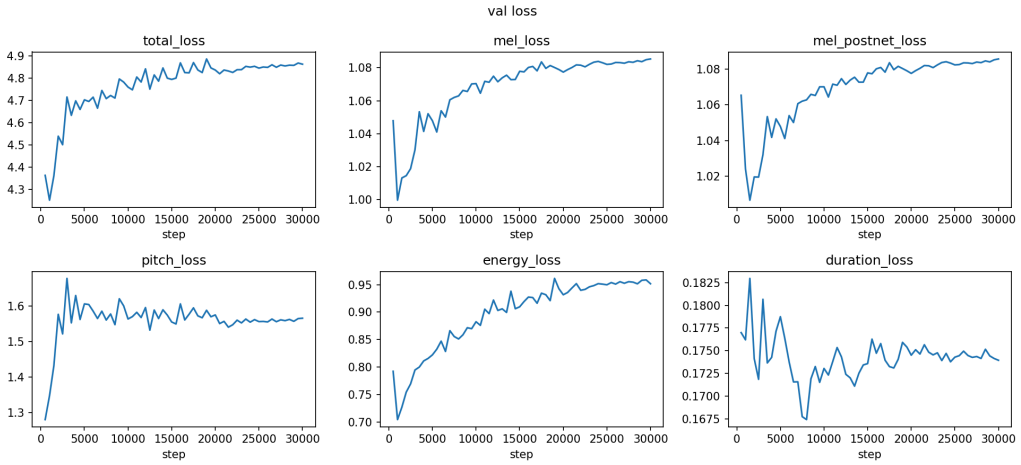


Fig.4. Validation Loss for Telugu

5. Observations on the generated speech

English. For all five English test sentences, the speech is not decipherable without the transcript. Even with the transcript, only the start of each audio file is clearly audible and can be inferred; the rest sounds muffled. A common pattern is that the speech is clear up to a particular word, after which it becomes messy and garbled. Because validation loss was lowest at around step 3,500, I also listened to several saved checkpoints: the step-10,000 checkpoint sounds best, though its quality is still not good, and at higher steps the quality degrades and the audio becomes more muffled. As a listener, I find it very hard to understand what is said without a transcript, because many words run into each other.

Telugu. All five Telugu test sentences are clear and decipherable. As a non-native Telugu speaker, I can judge only how the audio sounds, not what is being said. The audio at the reported (step-30,000) checkpoint sounds very clear and of good quality. From the step-5,000 checkpoint onward, each later checkpoint sounds better in prosody and manner of speaking.

These are informal impressions from a single listener, not a controlled listening test. The table gives frontend facts and word-level listening notes for the step-30,000 checkpoint.

ID	Frontend observation	Listening note (step 30,000)
E1	Long word became AE2 N T AY0 D IH0 S AH0 S T AE2 B L IH0 SH M AH0 N T EH1 R IY0 AH0 N IH0 Z AH0 M (lexicon or g2p_en, not logged)	Starts well up to the long word "antidisestablishmentarianism", after which the quality degrades.
E2a	conduct (noun): K AA1 N D AH0 K T	Listenable up to "orchestra", after which the quality becomes poor.
E2b	conduct (verb) got the same phones because lookup ignores context; verb form should be K AH0 N D AH1 K T	"Researchers" and "conduct" are understood; the rest is not quite understood.
E3	"rhyth m" kept as written: R AY1 DH + EH1 M, not rhythm (R IH1 DH AH0 M)	"Rhythm" is audible and "orchestra" is somewhat audible; the rest is muffled.
E4	10:45 AM and 12 November 2026 hand-expanded to words; andaroo = AE0 N D AA1 R OW0	Only the timing is understandable; the rest gets mixed up and is ultimately unclear.
T1	57 units; 4 words built by the table rule (న్యూరోఫార్మకోలాజికల్, పరిశోధనలో, ఔషధాల, విశ్లేషించారు)	The audio is clear and has a rhythm.
T2a	27 units; తాడుపై built by rule; పట్టు gets the same phones as the start of the T2b compound	The audio is clear and has a rhythm.
T2b	32 units; పట్టుచీరను and వేడుకకు built by rule; the model cannot distinguish the two senses of పట్టు	The audio is clear and has a rhythm.
T3	27 units; నిశ్శబ్దంగా built by rule	The audio is clear and has a rhythm.
T4	96 units; date/time hand-expanded to Telugu number words; six words built by rule; comma becomes sp	The audio is clear and has a rhythm.

In short: English is hard to follow without the transcript: the opening is understandable, then words blur together. Telugu is clear at the reported checkpoint and improves with training, so the quality problem is specific to the English model.

6. Variance-adaptor analysis

What the plots show. For each phoneme the variance adaptor predicts how long it is held, how high the voice is and how loud it is. Together these are the model's plan for how the sentence is spoken, and the decoder uses that plan to build the mel spectrogram. They are not the spectrogram itself.

Duration (bars): frames per phoneme, so the bars set rhythm and sentence length. **Pitch** (line, Hz): one F0 value per phoneme, hence flat steps. **Energy** (line): a loudness proxy per phoneme.

How the plots were made. inference.py runs each test sentence through the model once and records the values the adaptor feeds to the decoder: the rounded duration per phoneme (round(exp(log d) - 1) frames, one frame about 11.6 ms), and the per-phoneme pitch and energy predictions, de-normalised with the training mean and standard deviation from stats.json (pitch in Hz). Pitch and energy are drawn by repeating each phoneme's value over its predicted frames, with grey lines at phoneme boundaries. English plots used speaker id 0. They show the model's plan, not measurements of the audio.

English test sentences

E1 (very long word). Durations are mostly 3 to 8 frames. The long word (28 phones, about a third of the sentence) has no unusually long segments; the longest bars are on later words (EY1 of gained 14, IY0 of parliamentary 15). Pitch (about 137 to 198 Hz) peaks mid-word at SH M and dips to about 137 Hz on the final -ism, a normal reduction at an unstressed ending. Energy spikes to about 55 on the unstressed AY0 near the start of the word and falls to about 3 to 5 on -ism. The primary-stressed EH1 (-ar-) is not emphasised, which is atypical; a rare word probably receives generic values. The low-energy ending lies in the early-middle of the sentence, so it is a candidate contributor to blur there, not a demonstrated one.

E2a. Durations are 2 to 15 frames; the longest are the M of ceremony (14), EY1 of praised (14) and its final D (15). Pitch (about 139 to 205 Hz) is highest at the onset of conduct and falls through the word, from about 190 to 145 Hz, then to about 143 Hz at the start of during: the expected stressed-noun peak followed by a fall. Energy rises and falls phone by phone and peaks (about 42) on the function word by, above praised (about 33), which is atypical. There is no long low-energy stretch.

E2b. Durations are 2 to 21 frames. The standout is the schwa of the before proposed (21 frames, about 240 ms), longer than any stressed vowel, with flat pitch (about 148 Hz) and very low energy (about 4); the final T of evaluate (16) is also long. Pitch (about 116 to 195 Hz) starts high and trends downward to its lowest value at the final pause, a normal statement fall. Energy peaks on researchers (about 37) and around to evaluate (about 41), then stays very low (about 1 to 8) for the last third. Conduct gets the same phones as in E2a, so the verb stress (conDUCT) cannot be marked; its pitch falls from about 170 to 135 Hz here, driven only by context. The long, quiet, flat the is a candidate contributor to unclear speech in the later part.

E3. Durations are 3 to 15 frames; the longest are the AY1 of rhyth (14), EY1 of changed (15) and the final pause (12). Pitch (about 148 to 199 Hz) starts at its maximum, falls to about 150 Hz and rises on changed (about 194 Hz), which suits a content word. Energy is very high on the broken word (AY1 about 47, EH1 of m about 50) and then mostly low (about 5 to 10), apart from a bump (about 18 to 24) on changed. The split rhyth m is read as two units, so the front is loud and the rest of the sentence quiet, which is atypical for ordinary stress. The low energy after the opening words is a candidate contributor to unclear speech.

E4. The longest English sequence; durations are 2 to 16 frames, longest on AY1 of time (16), SH of should (15) and S of six and OW1 of so (14). The comma becomes a pause. Pitch (about 129 to 241 Hz) peaks on will start (about 241 Hz), falls to about 129 Hz near arrive and rises to about 190 Hz at the very end; an end rise is unusual for a statement. Energy peaks on meeting (about 42) and on the unstressed AH0 of will (about 56), and drops to about 2 across the expanded number words, which become many short, quiet phones. The sentence's largest peak on a function word is atypical; content words would be expected to carry it.

Telugu test sentences

T1. Durations are 1 to 13 frames. Pitch starts high (about 300 to 315 Hz), then falls and stays mostly around 180 to 220 Hz; energy starts high (about 85 to 95), then is lower and variable. This is the usual high start and declination of a statement. Four rare words were built by the table rule, which may add pronunciation errors the plots cannot show.

T2a. Durations are 1 to 15 frames. Pitch rises to about 315 to 320 Hz, drops to about 185 to 205 Hz and rises locally near the end; energy starts near 100, then falls sharply with several low regions. This is substantial phone-level variation, not flat prosody.

T2b. Durations are 1 to 14 frames. Pitch rises from about 220 to 260 to 265 Hz, then falls to about 190 to 205 Hz with local rises; energy starts at about 80 to 85, then drops and fluctuates. The homograph పట్టు gets

the same phones as in T2a (grip versus silk), yet the contours differ with context: the model uses context for prosody but cannot choose the correct pronunciation or sense.

T3. Durations are 1 to 16 frames with several long segments. Pitch peaks early (about 320 to 325 Hz), settles around 180 to 230 Hz and falls to about 155 to 160 Hz at the end, a clear final lowering; energy starts near 90, has a pronounced low region and stays low at the end.

T4. The longest Telugu sequence (96 units); durations are 1 to 18 frames, mostly moderate. Pitch starts around 270 to 280 Hz, drops into a lower band with local rises and trends lower at the end, as expected over a long sentence; energy opens high, fluctuates lower and rises modestly late. The number words and the comma pause make the plot dense and hard to compare.

Match with the audio. The English and Telugu plots share the same general shape (a high start, then lower and fluctuating values), yet only the English audio sounds poor, so that shape does not explain the English quality. Word-level comparison with the listening notes is partial and approximate: in E3 the audible words (rhyth m) coincide with the highest predicted energy (about 47 to 50) and the muffled remainder with low energy (about 5 to 10); in E1 the early energy spike falls where the long word begins and quality drops; in E2b the long, quiet, flat the lies inside the unclear part; E2a and E4 show no clear correspondence. Low predicted energy on consonant clusters is normal, so low values alone are not evidence of a fault. The plots are compatible with, but do not establish, a cause.

7. English and Telugu compared

Aspect	English	Telugu
Data	Indian-accented, 114 proxies, noisy, up to 11 s; 5.65 h	One female speaker, up to 20 s, 6.00 h
Units / frontend	ARPAbet with stress; lexicon + g2p_en	Grapheme/cluster units; table-based longest-match rule
Aligner	MFA english_us_arpa; 3,978 of 3,984	IndicMFA + generated dictionary; 1,859 of 1,864
Validation (min / final)	3.1089 (3.5k) / 3.3666	4.2522 (1k) / 4.8623
Quality	Clear start, then garbled; not decipherable without transcript; best near step 10,000	Clear and decipherable; improves from step 5,000 to the reported checkpoint (non-native listener)

Limitations and failures

- Checkpoint choice: English validation loss is lowest near step 3,500, yet the audio is best near step 10,000 and degrades later (overfitting). Telugu validation loss rises after step 1,000 while the audio keeps improving, so validation loss alone is a poor guide to audio quality.
- English garbling is specific to English: Telugu uses the same architecture, pipeline and universal HiFi-GAN vocoder and sounds clear, so the vocoder is unlikely to be the main cause (a vocoder-only test was run for English and it sounded clean).
- English data and alignment (candidate causes, not tested): Svarah is an accented, multi-speaker speech-recognition corpus rather than clean TTS data; speaker labels are demographic proxies, so one label can cover several voices; and the alignments came from the US-English MFA model and ARPAbet dictionary, which may fit Indian-accented pronunciations poorly. Only 6 of 3,984 utterances failed to align, but successful alignment is not the same as accurate alignment. Telugu used a language-specific aligner and one clean speaker, which is consistent with, but does not prove, a data or alignment explanation.
- English frontend: heteronyms unresolved (E2), g2p_en guesses broken words (E3), numerals hand-normalised (E4), no word-boundary tokens.
- Telugu frontend: character-level units with no schwa, geminate or sandhi rules, one pronunciation per homograph (T2), hand-written number words, out-of-vocabulary test words.
- Only about 5 h of data per language; no speaker balancing in English.

Improvements: choose checkpoints by listening as well as validation loss; align English with an accent-matched aligner and dictionary (for example MFA english_india_mfa, which uses the MFA phone set and so needs a matching frontend); use cleaner, speaker-consistent English data or true speaker labels; context-aware g2p and a general number/date normaliser; word-boundary tokens; Telugu pronunciation rules; per-language vocoder fine-tuning; MOS or intelligibility evaluation.

In short: the two models behave differently. Telugu is clear and improves with training; English degrades after about step 10,000 and garbles words from mid-sentence. Architecture and vocoder are shared, so the English problem points to the data, speaker labels, alignment and frontend, and to checkpoint choice.

8. Reproducibility

Python 3.13, PyTorch 2.11.0, numpy 2.x, librosa, PyWorld, tgt 1.4.4, g2p_en (English), MFA 3.4.3 via micromamba; minimum versions in requirements.txt. Hardware: Kaggle T4. Vocoder weights (generator_universal.pth.tar) come from the FastSpeech2 README and are not in the repository.

English (run in English/):

```
python prepare_data.py
mfa model download acoustic english_us_arpa && mfa model download dictionary english_us_arpa
mfa align raw_data/English english_us_arpa english_us_arpa preprocessed_data/English/TextGrid --clean -j 4
python make_config.py && python preprocess.py config.yaml
python train.py -c config.yaml
python inference.py -c config.yaml --restore_step 30000 --speaker_id 0 --out ../Results/English
python plot_losses.py output/log/English ../Results/English
```

Telugu (run in Telugu/; place Telugu_Dictionary_g2g.txt and Telugu_Acoustic_Model.zip from the IndicMFA Telugu release in mfa_models/):

```
python prepare_data.py && python build_dict.py
mfa align raw_data/Telugu mfa_models/dict_extended.txt mfa_models/Telugu_Acoustic_Model.zip
preprocessed_data/Telugu/TextGrid --clean --beam 100 --retry_beam 400 -j 4
python build_symbols.py && python make_config.py && python preprocess.py config.yaml
python train.py -c config.yaml
python inference.py -c config.yaml --restore_step 30000 --out ../Results/Telugu
python plot_losses.py output/log/Telugu ../Results/Telugu
```

Configs: English/config.yaml, Telugu/config.yaml.

In short: everything runs from each language folder with the commands above. Only the vocoder weights and the MFA models are downloaded separately.

9. References

1. Y. Ren et al., FastSpeech 2, ICLR 2021.
2. ming024/FastSpeech2, github.com/ming024/FastSpeech2.
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4. M. McAuliffe et al., Montreal Forced Aligner, Interspeech 2017.
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7. AI4Bharat, IndicMFA Telugu release (Aug 2024), github.com/AI4Bharat/IndicMFA.